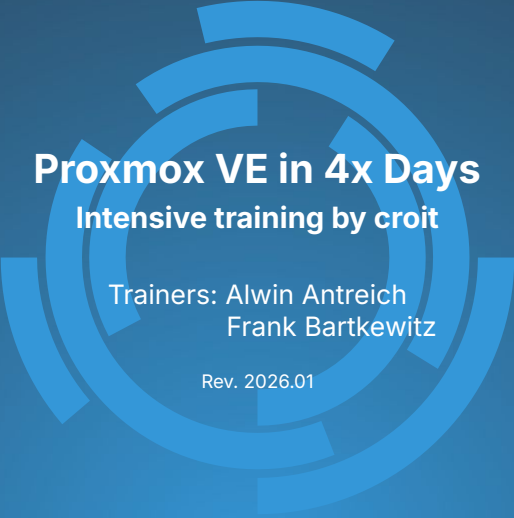




# **Proxmox VE in 4x Days**

**Intensive training by croit**

Trainers: Alwin Antreich  
Frank Bartkewitz

Rev. 2026.01

### **Alwin Antreich**

has been at [croit GmbH](#) as Storage & DevOps Engineer since 2021.

He worked as a software developer and trainer at Proxmox Server Solutions GmbH from 2017 to 2021.

He previously worked as a system and network administrator in various companies and can look back on 18 years of IT experience.



### **Frank Bartkewitz**

joined [croit GmbH](#) as Trainer & Consulting Specialist in 2025.

He worked as a Trainer for vpn solutions at secunet Security Networks AG from 2012 to 2024.

He previously served in German Armed Forces developing trainings for system administrators and in various other positions and can look back on 17 years of IT experience.



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- 4x full or 8x ½ days of training
- 45 minutes á session
- 15 minutes break between sessions
- 1 hour lunch break (full day only)
- Training material & Proxmox certificate will be send after the training

# Agenda - Day 1

- Introduction - croit GmbH & Training Lab
- About Proxmox & History of Proxmox VE
- Architecture & Concepts
- Single-node Setup
- Installation & Updates
- Proxmox VE Services
- Hardware Requirements
- Web-based GUI
- Command Line Interface
- Network Model
- Storage Model
- Storage, Network, Cabling
- Proxmox VE Guests

# Agenda - Day 2

- Full virtualization with KVM
- OS-level virtualization with LXC
- Migration: Bare-metal or virtual to VM
- VM Templates & Clones
- Proxmox VE Firewall
- User Management
- Basic Authentication
- Integrated Backup & Restore
- Storage Migration
- Virtual Appliances
- Guest Deployment
- PCI(e) passthrough
- Performance Baseline

# Agenda - Day 3

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- Multi-node Setup
- Cluster Architecture & Network Design
- Proxmox VE Clustering
- Proxmox VE Cluster Services
- pmxcfs - Advanced
- Clustering with QDevice
- Ceph - Installation, Setup, Management
- Security
- Updating a cluster
- Cluster-wide Backup
- Cluster-wide Firewall
- Performance Baseline

# Agenda - Day 4

- Hyper-Converged Infrastructure (HCI)
- User/Rights Management & Authentication
- Online/Live migration
- Certificate Management
- REST API
- Introduction to High Availability (HA)
- Hardware Requirements
- Proxmox VE HA Management
- Storage Replication Framework
- Advanced CLI
- Troubleshooting



(a) Martin Verges



(b) Matthias Grandl



(c) Andy Muthmann

Figure 1: The current owners of croit

Croit, founded in 2017, tackles the frustrations of managing Ceph storage. Combining the founders' deep expertise, we built a stellar team and a top-notch storage management solution. Our years of expertise led to a best-in-class storage management platform. Skip the years of struggle and grab control with Croit!

We provide comprehensive Proxmox 24/7 support, consulting, training, and integration services. Our certified experts tailor solutions to optimize your infrastructure, ensuring smooth migration, robust security, and maximized performance.

Since early 2021, croit began developing a new innovative solution to manage DAOS clusters. This solution extends the DAOS ecosystem and provides required connectivity and compatibility to enable everyone to use croit DAOS as their primary storage solution.

# Introduction - croit Storage Manager

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- Complete GUI-based cluster deployment and management
- Masters ZTP (zero touch provisioning)
- Includes live-booted OS (no installation of servers required)
- Automates Ceph and Linux upgrades (Opensuse x86\_64)
- Includes the ceph-mgr dashboard
- Also manages connectors (ceph-iscsi, SMB, RGW, NFS)
- Offers extensibility via REST API or custom event scripts

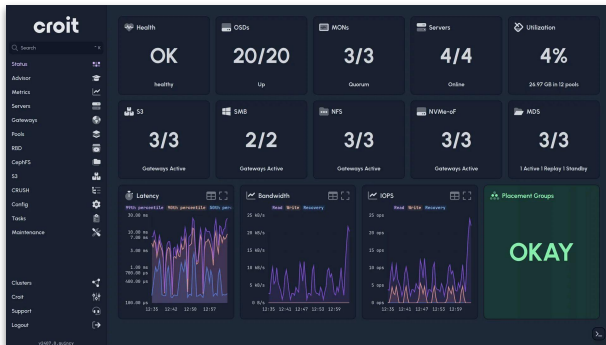


Figure 7: croit dashboard

To provide a test environment for every trainee the virtual lab is comprised of:

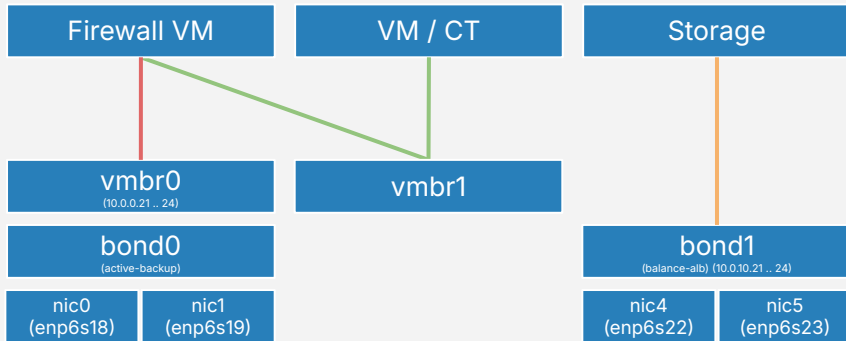
- A web-based Linux desktop
  - Your access point to the virtual lab
  - Please do all the work from there (including Google searches; easiest)
  - Use the flap  on the left to change the settings (e.g. image quality) and to access the copy-paste buffer
- 4x virtual servers
  - Multiple HDDs for storage simulation
  - Network connected to the management node within a private L2 VLAN
- A Windows server
- A Linux server

## Day 1 & 2 - Virtual infrastructure

**croit**

|                     |                                            | vl-srv1 .. 4                                                                | debian         | windows                                                              | NAS             |
|---------------------|--------------------------------------------|-----------------------------------------------------------------------------|----------------|----------------------------------------------------------------------|-----------------|
| User / Password:    |                                            | root / Croit!Croit!                                                         | root / Croit!  | Administrator / Croit!                                               | N/A             |
| Frontend<br>(bond0) | IP Address:<br>Gateway:<br>DNS:<br>Domain: | 10.0.0.21 .. .24 / 24<br>10.0.0.1<br>8.8.8.8 / 1.1.1.1<br>training.croit.io | 10.0.0.15 / 24 | 10.0.0.16 / 24<br>10.0.0.1<br>8.8.8.8 / 1.1.1.1<br>training.croit.io | 10.0.0.10 / 24  |
| Storage<br>(bond1)  | IP Address:                                | 10.0.10.21 .. 24 / 24                                                       |                |                                                                      | 10.0.10.10 / 24 |

# Day 1 & 2 - Host interface configuration



| Proxmox VE guests                                                                                                                                             |                                                                                                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| KVM                                                                                                                                                           | LXC                                                                                                                                                                                        |
| <ul style="list-style-type: none"><li>■ Linux,</li><li>■ 1x core</li><li>■ 2x nics (vmbr0 and vmbr1)</li><li>■ 1024 MB RAM</li><li>■ 4 GB disk size</li></ul> | <ul style="list-style-type: none"><li>■ 1x core</li><li>■ 1x nic (vmbr1)</li><li>■ 512 MB RAM</li><li>■ 4 GB rootfs</li><li>■ 2 GB mount point (separate)</li><li>■ unprivileged</li></ul> |

The documentation can be found at:

- [https://pve.proxmox.com/pve-docs/pve-admin-guide.html#qm\\_cloud\\_init](https://pve.proxmox.com/pve-docs/pve-admin-guide.html#qm_cloud_init)
- <https://cloudinit.readthedocs.io/en/latest/>

## Day 2 - Hook script

croit

```
root@vl-srv1:~# cat > /mnt/pve/samba/snippets/router-startup.sh
```

```
#!/usr/bin/bash
# chmod +x /mnt/pve/samba/snippets/router-startup.sh
# qm set 104 --hookscript samba:snippets/router-startup.sh
VMID="$1"
PHASE="$2"
NAME="ipfire"
if [[ -z $PHASE ]];
then
    echo "BAIL: not all arguments given!"
    exit 1;
fi
if [[ $PHASE == 'pre-start' ]];
then
    TARGET=$(pvesh get /cluster/resources --output-format json | jq ".[] | select(.name==\"$NAME\") | (.vmid)" -r)
    echo "${PHASE}: VM${TARGET}"
    NODE=$(pvesh get /cluster/resources --output-format json | \
        jq -r --argjson TARGET $TARGET '.[] | select(.vmid==$TARGET) | .node')
    STATE=$(pvesh get /cluster/resources --output-format json | \
        jq -r --argjson TARGET $TARGET '.[] | select(.vmid==$TARGET) | .status')
    if [[ $STATE == 'stopped' ]];
    then
        pvesh create /nodes/${NODE}/qemu/${TARGET}/status/start
    elif [[ $STATE == 'paused' ]];
    then
        pvesh create /nodes/${NODE}/qemu/${TARGET}/status/resume
    fi
fi
```

## Day 2 - Install Proxmox Backup Server

**croit**

The documentation can be found at:

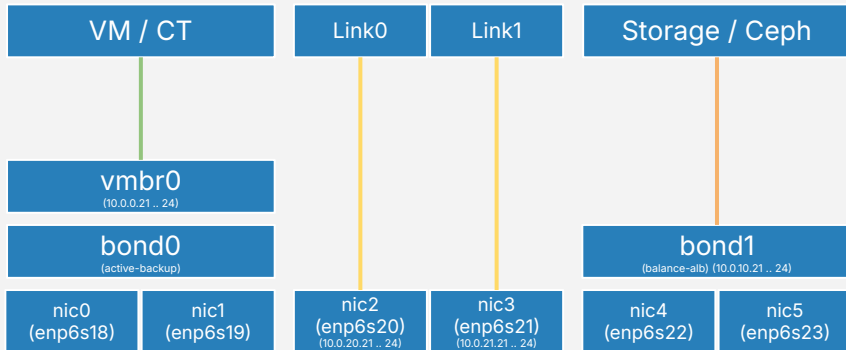
<https://pbs.proxmox.com/docs/introduction.html>

## Day 3 & 4 - Virtual infrastructure

**croit**

|                              |                                            | vl-srv1 .. 4                                                                | debian                                                               | windows                                                              | NAS             |
|------------------------------|--------------------------------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------|-----------------|
| User / Password:             |                                            | root / Croit!Croit!                                                         | root / Croit!                                                        | Administrator / Croit!                                               | N/A             |
| Frontend<br>(bond0)          | IP Address:<br>Gateway:<br>DNS:<br>Domain: | 10.0.0.21 .. .24 / 24<br>10.0.0.1<br>8.8.8.8 / 1.1.1.1<br>training.croit.io | 10.0.0.15 / 24<br>10.0.0.1<br>8.8.8.8 / 1.1.1.1<br>training.croit.io | 10.0.0.16 / 24<br>10.0.0.1<br>8.8.8.8 / 1.1.1.1<br>training.croit.io | 10.0.0.10 / 24  |
| Corosync 1<br>(nic2/enp6s20) | IP Address:                                | 10.0.20.21 .. 24 / 24                                                       |                                                                      |                                                                      |                 |
| Corosync 2<br>(nic3/enp6s21) | IP Address:                                | 10.0.21.21 .. 24 / 24                                                       |                                                                      |                                                                      |                 |
| Storage<br>(bond1)           | IP Address:                                | 10.0.10.21 .. 24 / 24                                                       |                                                                      |                                                                      | 10.0.10.10 / 24 |

## Day 3 & 4 - Host interface configuration



## Day 3 - Proxmox Auto-Installer

**croit**

The documentation of the auto-installer can be found at:

[https://pve.proxmox.com/wiki/Automated\\_Installation](https://pve.proxmox.com/wiki/Automated_Installation)

The http example server can be found at:

[https://pve.proxmox.com/wiki/Automated\\_Installation#Serving\\_Answer\\_Files\\_via\\_HTTP](https://pve.proxmox.com/wiki/Automated_Installation#Serving_Answer_Files_via_HTTP)

Copy answer file content into the *answer.toml*. (→)

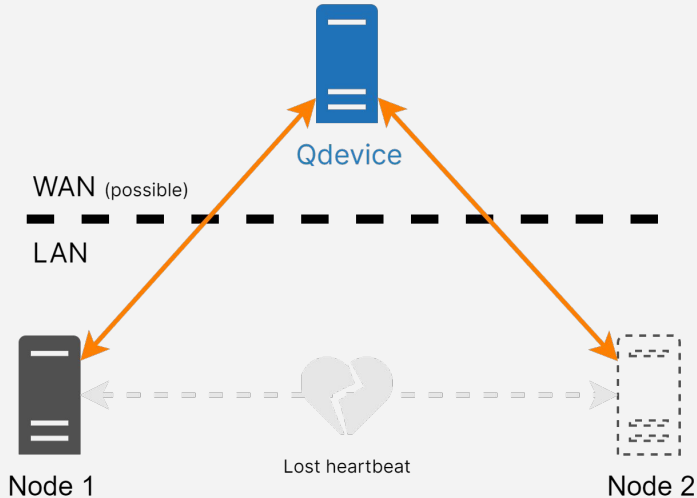
```
[global]
keyboard = "en-us"
country = "de"
fqdn.source = "from-dhcp"
fqdn.domain = "training.croit.io"
mailto = "mail@no.invalid"
timezone = "UTC"
root_password = "Croit!Croit!"

[network]
source = "from-dhcp"

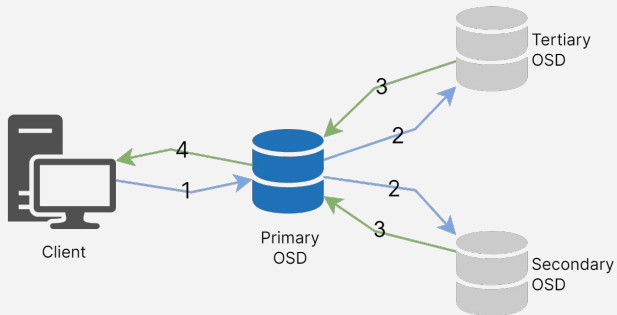
[network.interface-name-pinning]
enabled = true

[disk-setup]
filesystem = "ext4"
filter.ID_SERIAL_SHORT = "drive-scsi0"
```

## Day 3 - Two-node cluster



- cluster + public network: bond1
- 3x MON
- 3x MGR
- 8x OSD (2x each node; DB/WAL on separate SSD)
- 2x MDS (active + standby-replay)
- 1x pool: rbd (replicated)
- 1x CephFS

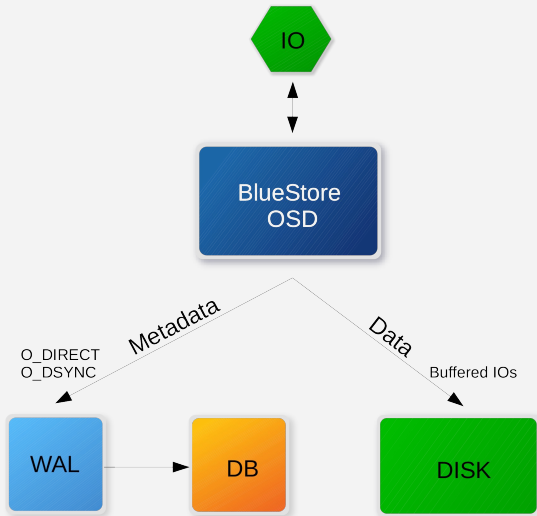


## Strong Consistency

CAP Theorem: CP System  
(Consistency / network partition)

1. A client writes to the primary OSD
2. Primary OSD sends data to the replica OSDs + writes data locally
3. Replica OSDs write data locally, and send the acknowledgement to the primary OSD
4. The primary OSD acknowledges completion to the client

# Bluestore internals



- **ceph osd set noout**
  - keep OSDs in the cluster, no recovery of data
  - ignores `mon_osd_down_out_interval` settings (default 10 minutes)
  - use `ceph osd unset noout` to revert the flag
- **ceph osd destroy <ID>**
  - keeps OSD in crush map
  - no extra rebalance of data (opposed to deleting an OSD)
- **ceph-volume lvm zap --destroy /dev/sd<X>**
  - clear any LVM or partition table
  - best first used on any replacement disk
- **ceph-volume lvm create --data /dev/sd<X> --osd-id <Y>**
  - re-create an OSD with the same OSD ID.

## Day 3 & 4 - Prioritizing recovery IO vs client IO (mClock)

**croit**

- **balanced (default)**
  - client: 50%
  - recovery: 50% (background)
- **high\_client\_ops**
  - client: 60%
  - recovery: 40% (background)
- **high\_recovery\_ops**
  - client: 30%
  - recovery: 70% (background)

```
~$ ceph config set osd osd_mclock_profile high_client_ops
~$ ceph config set osd.0 osd_mclock_profile high_recovery_ops
```

<https://docs.ceph.com/en/latest/rados/configuration/mclock-config-ref>

- **osd max backfills (default 1)**
  - Adjusts how many PGs per OSD are rebalanced in parallel
  - Increase this if the recovery is too slow
- **osd recovery sleep ssd/hybrid/hdd (default 0/0.025/0.1 seconds)**
  - ssd/hybrid/hdd refers to the type of the OSD,  
hybrid OSDs are HDDs with an SSD as journal or DB device
  - Injects a short sleep between each transferred object
  - Increase this to decrease recovery speed if client IO suffers

```
~$ ceph tell osd.* injectargs '--osd_max_backfills=4'
~$ ceph tell osd.* injectargs '--osd_recovery_sleep_hdd=0.2'
```

## Day 4 - Proxmox Datacenter Manager (Beta)

**croit**

Instructions can be found at: [https://pve.proxmox.com/wiki/Proxmox\\_Datacenter\\_Manager\\_Beta\\_Documentation](https://pve.proxmox.com/wiki/Proxmox_Datacenter_Manager_Beta_Documentation)

Roadmap: [https://pve.proxmox.com/wiki/Proxmox\\_Datacenter\\_Manager\\_Roadmap](https://pve.proxmox.com/wiki/Proxmox_Datacenter_Manager_Roadmap)

1. Add the PDM apt repository to your host.

```
~$ echo 'deb http://download.proxmox.com/debian/pdm trixie pdm-test' > /etc/apt/sources.list.d/pdm-test.list
```

2. Add the enterprise release key.

```
~$ wget https://enterprise.proxmox.com/debian/proxmox-release-trixie.gpg -O /etc/apt/trusted.gpg.d/proxmox-release-trixie.gpg
```

3. Install the packages.

```
~$ apt update
~$ apt install proxmox-datacenter-manager proxmox-datacenter-manager-ui
```

Visit: <https://10.0.0.100:8443>

# Thank you for your attention!

Are you looking for an exciting job in the  
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[www.croit.io](http://www.croit.io)  
[contact@croit.io](mailto:contact@croit.io)

